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Semicentenial dynamics of arable lands and fertility arable soils of the Republic of Tyva

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Abstract. The article presents an analysis of data within the fifty-year period on the dynamics arable lands and the main indicators of soil fertility of agricultural lands of the Republic of Tyva. The soil cover of the forest-steppe zone of the Republic is mainly represented by southern and usual chernozems, while the steppe zone is represented by chestnut soils. Arable land is located mainly on chestnut soils (65%) and chernozems (27%). On 1/1/2020 the arable land area is 135.5 thousand hectares. It is established that agricultural land for the period 1970–2020 decreased by 1.3 times, including arable land – by 3.6 times, hayfields – by 2 times, pastures – by 1.7 times. In the course of long-term agricultural use, the content of humus, mobile phosphorus and exchangeable potassium in the arable horizon changes, as well as the acidity of arable land. It was revealed that the humus state of arable soils in Tuva is estimated to be low and very low levels of fertility.

1. Introduction

World and national experience show that high and sustainable agricultural productivity is possible only if all agrochemical and ecological factors necessary for normal growth and development of plants, crop formation and quality, and prevention of land degradation are taken into account [1–4].

The territory of the Republic of Tuva occupies 16,860.4 thousand hectares, of which 3,363.9 thousand hectares are agricultural land. The soil cover is characterized by great diversity and sharp territorial heterogeneity. The nature of the soil fully reflects the uniqueness of natural and climatic conditions and emphasizes the geographical originality of the territory. The climate is sharply continental, with dry and hot summers, long cold winters, annual precipitation in the hollows of 150–350 mm, most of them fall during the warm period with a maximum in late July – early August. The snow cover is insignificant – 10 cm. Most of the snow is blown away from open areas. The soil freezes deeply and thaws completely only in late June and early July. Late spring and early autumn frosts are typical. The duration of the period with temperatures above 10 °C is 100–125 days, the sum of temperatures for this period is 1,500–2,100 °C [5, 6].

At present, in Tuva, the widespread deflation processes in intermountain hollows, which are largely due to the dry climate, strong insolation, lack of land cover in agricultural areas, and the ploughing of primitive chestnut soils with a poorly developed soil horizon, with a close occurrence of the soil-forming underlying rock, are the cause of deep degradation of soil fertility. Due to the mountainous relief, light granulometric composition of soils, and inefficient use of 2,034 thousand hectares of land, 60.5% of deflated arable lands in Tuva is made up of chestnut soils, which contain only 0.13–2.52% of humus [7, 8], while in Western Siberia this figure is in the range of 1.5–4.0% [9], and in the European part of Russia – 2.3–3.3% [10].

The purpose of the work is to analyze the dynamics of arable lands and the main indicators of agricultural soil fertility in the Republic of Tuva for 50 years and identify emerging trends.

2. Subject and method of research

Mountain territories of Tuva, with their various mountain soils, occupy 14,011 thousand hectares, or 82.2% of the total land area, flat (hollow) territories – 3,039 thousand hectares, or 17.8%. The composition of steppe and desert-steppe lowlands (basin) soils are dominated by types of automorphic series. Together with genetically similar soils with slightly increased moisture content, they occupy 58% of the area of steppe plains. Semihydromorphic soils occupy 34%, hydromorphic soils and solonchaks – up to 5%, about 3% of the area are covered by sands [11].

The area of chernozems is 151 thousand ha, meadow-chernozem soils – 134 thousand ha, mountain chernozem – 341 thousand ha, chestnut soils – 1,195 thousand ha, meadow-chestnut soils – 17 thousand ha, mountain chestnut soils – 1,454 thousand ha, brown desert-steppe and brown meadow-desert-steppe soils – 18 thousand ha.

On 1/1/2020, the share of farmlands in the structure of agricultural lands on the Republic are 19.9%, of which more than 70% of arable lands falls on the share of Tandinsky, Sut-Kholsky, Piy-Khemsky and Kaa-Khemsky kozhuuns [12].

To assess the dynamics of arable soils fertility in Tuva, statistical materials of the Federal service for state registration, cadastre and cartography for the Republic of Tuva were used [13]. To characterize the fertility of arable soils, we used materials from the Federal state budgetary institution of the agricultural chemical service "Tuvan", and author's materials obtained as a result of laying soil sections, describing morphological features, and selecting soil samples according to generally accepted methods. Chemical analyses of soils were performed according to generally accepted methods in soil science [14], water-chemical properties were determined by methods accepted in soil physics [15]. Statistical data processing was performed using the Statistica 8.0. program.

3. Results and discussion

In the development of agriculture in Tuva during the Soviet period, there was a tendency to increase the acreage, and in 1970 it reached 493.1 thousand hectares, which was due to the implementation of the plan of the XXIV Congress of the CPSU. Over the next decade, there is a further increase in arable lands [16–18].

Before the beginning of the land reform in Russia, the agricultural land use of the Republic was characterized by the activities of large state farms (72 state farms on 1/1/1991) [19]. Agricultural production was carried out according to zonal farming systems on 3,573.6 thousand hectares of agricultural lands (table 1). In 1990, the arable area was 429.5 thousand hectares, of which 99% was in state farms. Due to the development of large areas of virgin lands that are unstable to deflation, the natural balance of the region has been disrupted. Soil degradation and land desertification have increased on arable lands.

During the period of reforms in the 1990s, negative trends in the country's economy had a more acute impact on the socio-economic development of the Republic, given the complex of unresolved social problems, as well as the relative transport isolation, which is the only region in Russia that does not have a railway connection with the rest of the country. Tuva's agriculture, without state support, began to deteriorate rapidly. Agricultural lands turned out to be unclaimed and, gradually overgrown with weeds, turned into the fallow lands. When large farms were reformed, the role of Arat farms (farmers') was strengthened. However, the main areas of farmlands still remain in the use of state farms and agricultural production cooperatives (more than 70%), which presented a problem in the land use of the Republic and required legislative regulation.

Table 1. Agricultural lands of Tuva, thousand hectares.

Farmlands	1970	1980	1990	1995	2000	2005	2010	2015	2020
Agricultural land, including:	4,923.8	4,610.0	6,812.2	6,624.2	3,161.9	1,219.9	3,371.7	3,367.3	3,363.9
croplands, incl.:	4,623.2	3,692.0	3,573.6	3,116.7	2,491.9	855.8	2,661.3	2,656.8	2,653.6
arable lands	493.1	416.0	429.5	361.1	155.7	51.3	135.6	135.5	135.5
including irrigated	-	-	37.4	32.2	25.6	11.7	15.1	15.1	15.1
fallow lands	5.7	-	1.3	1.6	49.6	7.3	61.4	61.4	61.4
hayfields	107.5	72.0	56.0	48.9	53.2	18.4	54.8	54.8	54.8
pastures	4,016.6	3,204.0	3,086.8	2,705.1	2,233.2	778.7	2,409.3	2,404.9	2,401.9

In the new market conditions in Tuva, there is an even greater decrease in the arable lands of grain crops. Thus, in 2000, compared to 1980, the arable lands of grain crops decreased by 8.4 times, wheat by 6.7 times, and forage crops by 11 times. A sharp decrease in the area of agricultural land occurred before 2005 by 1,559.9 thousand hectares. These lands were transferred to the lands of the forest Fund, specially protected natural territories, state land reserves and fallow lands. At the present time, there is a slight increase in them, especially in hayfields and pastures [20, 21].

During the analyzed fifty-year period, agricultural lands decreased by almost 2 times, including arable lands – by 3.6 times, hayfields – by 2 times, pastures – by 1.7 times. On 1/1/2020, agricultural lands in Tuva occupy 2,653.6 thousand hectares or 15.7% of the total area of the region. The share of arable land in the structure of agricultural lands in the current period is 5.1%, hayfields – 2.0%, pastures – 90.5%, fallow – 2.3%. More than 60% of arable lands are concentrated in the more favorable Central Tuva basin (Tandinsky, Sut-Kholsky, Piy-khemsky and Kaa-khemsky kozhuun), of which about 11% is irrigated [12]. Among the subjects of the Siberian Federal district, Tuva has the lowest availability of arable land – 0.2 ha per capita. For comparison, we note that the availability of arable land in the Altai region reaches 2.5 ha, the Novosibirsk region – 1.4 ha, Khakassia – 1.1 ha, and the Krasnoyarsk region – 1.0 ha [22].

The results of the study of the dynamics of soil fertility in Tuva showed that the agricultural soils of the Republic as a whole are insufficiently provided with humus and basic nutrients. The area of arable lands with a very low humus content in the soil is 79%, with an average – 15% of the total area; phosphorus with a very low and low content – 63%, with an average – 27% of the total area; potassium with a very low and low content – 65%, with an average – 29% of the total area. Arable soils of Tuva are located mainly on chestnut soils (69%) and black soils (27%) [2]. The main properties of these soils are given according to our data from typical soil sections (table 2).

The granulometric composition of ordinary chernozems is diverse: from loamy to sandy. Ordinary chernozems have a low density of addition (0.98–1.11 g/cm^3). The distribution of humus along the profile is characterized by a sharp concentration in the upper layer and a subsequent very rapid decline with depth. This is due to the fact that the majority of plant roots are located in the upper layers of the soil. The cation exchange capacity varies in the range of 30–40 mg-equiv./100 g, depending on the humus content and granulometric composition. The composition of the CCE is dominated by exchange Ca^{2+}. Southern chernozems, unlike ordinary ones, are characterized by a lower thickness of the humus-accumulative horizon, intensive mineralization of organic matter, and sandy loam varieties are more common among them. The reaction of the medium within the profile varies from neutral to slightly alkaline. The soil-absorbing complex is completely saturated with base cations, among which dominates Ca^{2+} and a small amount of exchange Na^{+} is present. These soils are characterized by low availability of mobile forms of nitrogen, medium and high – mobile forms of phosphorus and potassium.

Table 2. The main indicators of arable soils fertility in Tuva in the soil layer 0–20 cm.

Soil	Humus, %	Nitrogen, %	Easily hydrol. nitrogen, mg/kg	P_2O_5, mg/kg	K_2O, mg/kg	CCE, mg-equiv./100 g	pH_{H2O}	D, g/cm^3	Content of particles <0.01 mm, %
Chernozem usually light loamy	5.61	0.31	154	24	213	36	7.6	0.99	24.5
Chernozem southern sandy loam	4.18	0.27	133	21	200	20	7.5	1.06	14.6
Dark chestnut medium loamy	3.3	0.24	138	20	190	30	7.2	1.19	32.5
Chestnut sandy loam	1.66	0.14	64	19	159	14	7.4	1,29	13.2
Light chestnut sandy loam	1.11	0.10	82	19	116	16	7.4	1.29	12.5

Chestnut soils were formed in the dry steppes of the republic. There are three subtypes of the chestnut soil type: dark chestnut, chestnut, and light chestnut. Soil subtypes reflect the increasing aridity of the bioclimatic regime. Dark chestnut soil is similar in some properties to southern chernozem. Chestnut and, especially, light chestnut, as the most xeromorphic subtypes, differ in shortening of the accumulative-humus horizon, increased density, unsatisfactory structural state (22–32% of water-bearing aggregates against 48–78% in dark chestnut soils). Chestnut soils are characterized by sandy loam and sandy, rarely light loam granulometric composition, slightly alkaline and alkaline pH. Humus reserves decrease from dark chestnut (72 t/ha) to light chestnut (46 t/ha) [7].

The humus state of the soils of the agricultural territory of the republic is estimated mainly by low and very low levels of fertility. According to the data obtained [8], according to the content of humus, agricultural soils are arranged in the following decreasing series: usually chernozem > southern chernozem > dark chestnut > chestnut > light chestnut (table 3).

Table 3. Statistical parameters of humus content in the arable soil layer of the agricultural territory of Tuva, %.

Soil	n[a]	min-max[b]	X[c] ± Sx[d]	S_2[e]	V[f]
Chernozem usually	15	2.63–7.12	4.48±0.33	1.62	28
Dark chestnut	38	2.35–6.73	3.97±0.15	0.87	23
Dark chestnut	46	2.27–4.85	3.44±0.10	0.43	19
Chestnut	76	1.43–4.50	2.65±0.11	1.03	38
Light chestnut	27	1.42–3.82	2.18±0.15	0.59	35
Alluvial turf stepped	45	1.313.69	2.40±0.09	0.40	26

[a] n – sample size.
[b] min – minimum value, max – maximum value.
[c] X – average value.
[d] Sx – error of the average.
[e] S_2 – variance.
[f] V – coefficient of variation.

The reasons for the low soil fertility of Tuva are the light granulometric composition, less intake of root and crop residues into the soil compared to natural vegetation, deflation processes that sharply increased after plowing virgin lands, high intensity of humus mineralization processes in a hot, dry climate on the agricultural territory of the republic.

4. Conclusion

The generalized results of the dynamics of arable lands and the main indicators of arable soils fertility in the Republic of Tuva over 50 years showed that as a result of the reform of the Russian agro-industrial complex, the share of agricultural land in Tuva has decreased by 1.3 times since 1970. The decline in agricultural activity that occurred after 1990 was accompanied by a significant reduction in the area of arable land and its conversion to fallow land. The decline in agricultural land occurred until 2005. At present, there is a slight increase in them, especially in hayfields and pastures. New trends in land use in the republic require an assessment of the state and nature of agrogenic transformation of soils. It is revealed that the arable soils of the republic as a whole are insufficiently provided with humus and basic elements of nutrition. The area of arable land with a very low humus content in the soil is 79%, with an average – 15% of the total area; phosphorus with a very low and low content – 63%, with an average – 27% of the total area; potassium with a very low and low content – 65%, with an average – 29% of the total area. The fertility of arable soils is low. To preserve soil fertility, it is necessary to apply high doses of organic and mineral fertilizers, grow sideral crops, and also monitor the soil, control the rational use of natural and land resources.

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